U.S. NATIONAL SECURITY ENCLAVE, SPECIAL ACCESS PROGRAM (SAP), SCIF/TEMPEST, CRYPTOGRAPHIC & FORENSIC ENGINEERING ARCHITECTURE

Exhaustive Primary-Source Technical, Statutory, Control-Schema, and Causal-Dependency Audit of All 265 Classified Infrastructure Terms

Audit Basis & Evidentiary Standard: Compiled exclusively from immutable statutory, executive, military, federal regulatory, and cryptographic/hardware engineering records. Zero reliance on secondary commentary, commercial advertising, or informal summaries. Every operational state transition—from initial classification by an Original Classification Authority (OCA) to physical/electromagnetic containment inside an ICD 705 SCIF/SAPF, cryptographic encapsulation across HAIPE Type 1 network boundaries, kernel-level eBPF/XDP telemetry verification, and terminal NIST SP 800-88 / NSA/CSS EPL 9-12 media destruction—is mapped as a deterministic causal chain.

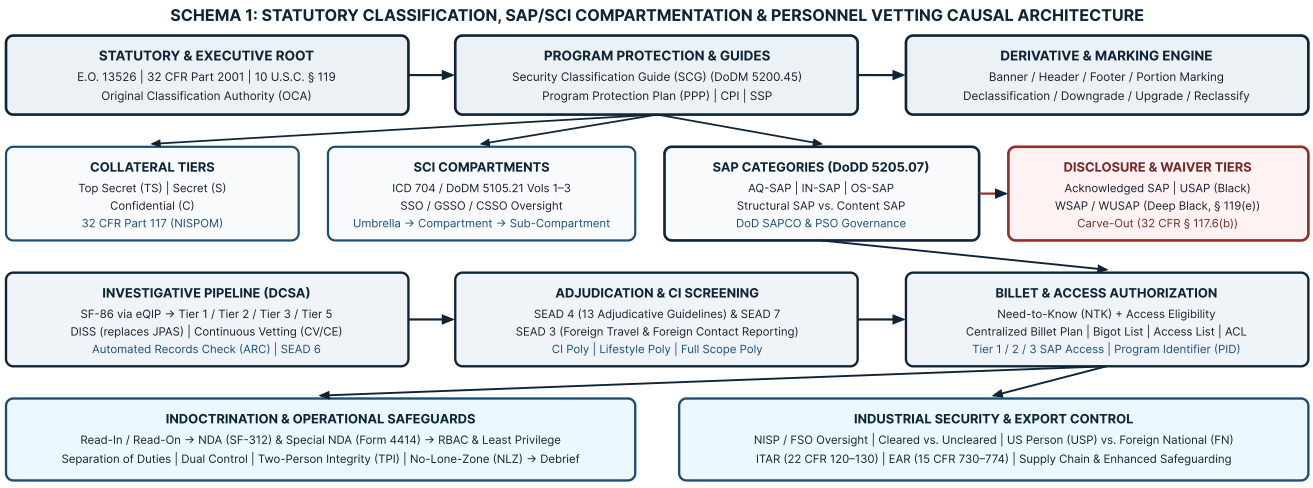
Architectural Domain	Primary Statutory, Military & Standards Baselines	Core Causal & Engineering Mechanisms
Domain I: Governance, OCA, SAP/SCI & Vetting	E.O. 13526; 10 U.S.C. § 119; 32 CFR Part 2001; 32 CFR Part 117 (NISPOM); DoDD 5205.07; DoDM 5205.07; ICD 704; SEAD 3/4/6/7; 22 CFR ITAR; 15 CFR EAR	OCA damage determinations (TS/S/C) → SCG/PPP/CPI → AQ-SAP, IN-SAP, OS-SAP (Acknowledged, USAP/Black, WSAP/WUSAP/Deep Black, Carve-Out) → SF-86/eQIP Tier 1–5 & DISS/CV/CE/ARC → Polygraph → Billet/Bigot List → Read-In, NDA, TPI & NLZ.
Domain II: SCIF/SAPF Physical & TEMPEST/EMSEC	ICD 705; IC Tech Spec v1.5.1; UL 2050; AA-F-358K / FED-STD-809D; FF-L-2740B; NSTISSAM TEMPEST/1-92; MIL-STD-188-124B; FCD 1/2 COOP; 36 CFR Part 1234	STC 45/50 acoustic barriers; GSA Class 5 (10-min force) & Class 6 containers; FIPS 201-3 PACS mantraps; UL 2050 Extent 3 IDS (BMS, PIR/Microwave, Tamper); ≥100 dB RF shielding & MIL-PRF-15733 power/data filters; Red/Black spatial & optical isolation.
Domain III: Classified Networks, Cloud & Crypto	CNSSI 1253; CNSSI 4009; DISA CC SRG; DoD Cloud Playbook; HSPD-12 / FIPS 201-3; FIPS 140-2 / 140-3; ISO/IEC 15408 CC; NSA CNSA 2.0; DoDI 8523.01; TM 11-5810-292-13&P	JWICS & NSANet (TS/SCI) / SIPRNet (Secret IL6) / NIPRNet (IL2–IL5); C2S/C2E (AWS Secret/Top Secret, Azure Gov IL5/IL6, Google, Oracle); DISA BCAP → VDSS → VDMS → TCCM; CAC/PIV PKI & HSM; Type 1 HAIPE (KG-175 TACLANE) & KMI fill devices (SKL, AN/CYZ-10, KOI-18).
Domain IV: Kernel Telemetry & Media Sanitization	IETF RFC 9669 (BPF ISA); Linux eBPF Verifier; XDP / AF_XDP; NIST SP 800-86; NIST SP 800-88 Rev. 1 & Rev. 2; NSA/CSS Manual 9-12; INCITS 529 ACS-4; NVMe Base Spec	64-bit eBPF ISA & static verifier; XDP driver-level packet filtering; LSM BPF (`id`=29) & lockless ringbuf (`id`=27) anti-forensic telemetry; NIST SP 800-86 Chain of Custody (CoC); NIST 800-88 Clear/Purge/Destroy (ATA 0xF4/0xB4, NVMe 0×80/0×84, CE/Zeroization, ≥5000 Oe Degauss, ≤1×5 mm Shred) → COD.

Document Identifier: ENG-AUDIT-9TXT-2026-REV1 | **Taxonomy Coverage:** 265 / 265 Unique Terms (100% Verified) | **Compilation Date:** November 2026

1. Statutory Classification Authority, Program Protection, SAP/SCI Compartmentation, and Personnel Vetting Architecture

The legal, administrative, and operational architecture of United States classified information systems originates from a strict constitutional and statutory chain of delegation. Under Executive Order 13526 (Classified National Security Information) and its implementing directive 32 CFR Part 2001, information may only be originally classified by a designated **Original Classification Authority (OCA)**—the President, the Vice President, agency heads, or officials delegated authority in writing by position under E.O. 13526 § 1.3. When an OCA determines that unauthorized disclosure of information falling within one of the eight mandatory categories of E.O. 13526 § 1.4(a)–(h) could reasonably be expected to cause damage to national security, the OCA assigns one of three hierarchical **Classification** levels: **Top Secret (TS)** for exceptionally grave damage, **Secret (S)** for serious damage, or **Confidential (C)** for identifiable damage. Standard clearance eligibility at these three levels without special compartmentation overlays is designated as **Collateral** security (DoDM 5200.01 Volume 1).

To propagate original classification decisions deterministically across government commands and defense industrial contractors, the OCA issues a **Security Classification Guide (SCG)** (DoDM 5200.45) reviewed at minimum every five years (32 CFR § 2001.16). Any cleared specialist or engineer who incorporates, paraphrases, restates, or generates new material from an SCG or source document acts as a **Derivative Classifier** (**Derivative** / **Classifier**) and must carry forward the exact classification rules, duration, and declassification instructions (32 CFR § 2001.22; ICD 703). Every classified **Document** (whether electronic or hardcopy **Paper**) requires standardized **Marking**: top and bottom centered **Banner** lines (situated in the page **Header** and **Footer**) specifying the highest overall classification and control caveats, paired with parenthetical **Portion Marking** (**Portion**, e.g., `(U)`, `(C)`, `(S)`, `(TS//SI//TK)`) preceding every title, heading, paragraph, table, and figure caption so downstream derivative classifiers can isolate the sensitivity of individual elements. Lifecycle transitions are governed by explicit administrative actions: automatic or mandatory review **Declassification** (**Declass**, typically at 10, 25, or 50/75 years under E.O. 13526 Part 3), scheduled or event-driven **Downgrading** (**Downgrade**) to a lower classification tier, **Upgrading** (**Upgrade**) to a higher classification tier when combined facts create heightened risk, and **Reclassifying** (**Reclassify**) previously declassified material under the personal written authority of an agency head when physical recovery remains feasible (E.O. 13526 § 1.7(c)).



Schema 1: End-to-End Statutory Delegation, SAP/SCI Compartmentation, DCSA Personnel Vetting, and Operational Access Control Causal Graph.

1.1 Program Protection, Risk Assessment, and Multidisciplinary Countermeasures

Within Department of Defense acquisition programs (DoDI 5200.39; DoDI 5000.83; DoDI 5000.90), engineering protection begins by identifying **Critical Program Information (CPI)**—hardware, software, algorithms, manufacturing processes, or mission data whose compromise would degrade combat effectiveness or shorten the operational life of a weapon or intelligence system. To defend CPI across the acquisition lifecycle, the Program Manager and **Program Security Officer (PSO)** establish a milestone-gated **Program Protection Plan (PPP)** (**Program Protection**). At the information system level, the Authorizing Official (AO) requires a formal **System Security Plan (SSP)** (**Security Plan**) under NIST SP 800-18 Rev. 2 and NIST SP 800-53 Rev. 5 control PL-2 (or the Joint Special Access Program Implementation Guide, JSIG).

Both the PPP and SSP are driven by a formal **Risk Assessment** (NIST SP 800-30 Rev. 1) evaluating risk as a causal function of an adversary **Threat** source exploiting an architectural or procedural **Vulnerability** to produce a quantifiable mission **Impact**. Risk **Mitigation** is achieved by engineering a layered **Control**, **Safeguard**, and **Countermeasure** (**Countermeasures**) baseline spanning five mandatory security disciplines: **Technical** controls (cryptography, hardware roots of trust, kernel mandatory access control), **Administrative** controls (SCGs, SOPs, NDAs, security plans), **Physical** controls (SCIF/SAPF barriers, GSA containers, PACS, IDS, guard forces), **Personnel** controls (Tier 3/5 vetting, Continuous Vetting, polygraph, SEAD 3 reporting), and **Industrial** controls (facility clearances, FOCI mitigation, DD Form 254 contractual binding).

1.2 Compartmented Topologies: SCI and Special Access Programs (SAP)

When standard Collateral safeguards cannot adequately protect intelligence sources, methods, or advanced capabilities, E.O. 13526 § 4.3 authorizes **Compartmented** control regimes enforcing **Enhanced Safeguarding** and strict access barriers:

- **Sensitive Compartmented Information (SCI):** Governed by the Director of National Intelligence under ICD 704 and implemented across the DoD by DoDM 5105.21 Volumes 1–3, SCI protects national intelligence sources, methods, and analytical processes (most commonly at the **TS/SCI** level). SCI is organized hierarchically under an overarching control system or **Umbrella**, subdivided into a functional **Compartment** (e.g., Special Intelligence `SI`, Talent Keyhole `TK`, Klondike `KDK`, `HCS`), and further partitioned into a granular **Sub-Compartment** (and projects) to isolate specific technical sensors or collection operations.
- **Special Access Program (SAP):** Governed by 10 U.S.C. § 119, DoDD 5205.07, DoDI 5205.11, and DoDM 5205.07 Volumes 1–4, every DoD SAP is categorized by mission into one of three functional classes: an **Acquisition SAP (AQ-SAP)** protecting research, development, test, evaluation, procurement, or modification of weapon platforms; an **Intelligence SAP (IN-SAP)** protecting sensitive intelligence or counterintelligence operations; or an **Operations and Support SAP (OS-SAP)** protecting sensitive military operational plans, deployments, and support activities. Architecturally, SAPs are also classified into a **Structural SAP** (which conceals the organizational structure, budget line, and relationship of the program) or a **Content SAP** (which protects specific technical parameters or mission payload data while the surrounding organizational wrapper remains visible).
- **SAP Disclosure Tiers and Congressional Reporting Gates (Acknowledged, USAP/Black, WSAP/WUSAP/Deep Black, and Carve-Out):** An **Acknowledged SAP** has its existence publicly or officially acknowledged while its technical details remain compartmented. Conversely, an **Unacknowledged SAP (USAP)**—operationally termed a **Black** or **Black Program**—has its very existence protected as classified information across all unclassified and collateral networks; under 10 U.S.C. § 119(d), no funds may be obligated for a USAP until 30 days after the Secretary of Defense notifies the congressional defense committees. When the Secretary of Defense personally determines in writing that inclusion of a USAP in standard SAP congressional reports would adversely affect national security (10

U.S.C. § 119(e)), the program is designated a **Waived SAP (WSAP)** or **Waived Unacknowledged SAP (WUSAP)**—colloquially termed **Deep Black** (**Waived**)—for which reporting is legally restricted to the eight Chairs and Ranking Minority Members of the Senate and House Armed Services and Appropriations Defense Subcommittees (10 U.S.C. § 119(g)). Furthermore, under 32 CFR § 117.6(b) and DoDD 5205.07, the Secretary of Defense may designate a SAP as a **Carve-Out** contract, relieving the Defense Counterintelligence and Security Agency (DCSA) of routine industrial security inspection authority and vesting exclusive security oversight directly in the SAP Central Office (SAPCO) and PSO.

Classification / SAP Category	Statutory & Regulatory Authority	Oversight & Accreditation Chain	Access, Vetting & Disclosure Invariants
Collateral: Top Secret (TS), Secret (S), Confidential (C)	E.O. 13526 § 1.2; 32 CFR Part 2001; DoDM 5200.01; 32 CFR Part 117 (NISPOM)	Command **Security Officer** (Gov) / **FSO** & **DCSA** (**DSS** legacy) for Industrial NISP facilities	Tier 5 (TS) or Tier 3 (S/C) + CV/CE + SF-312 **NDA** + verified **Need-to-Know (NTK)**
Sensitive Compartmented Information (SCI / TS/SCI)	ICD 704; ICD 705; DoDM 5105.21 Vols 1–3	**DIA** / **NSA** / **NRO** / **NGA** CSA; local **SSO**, **GSSO**, and Contractor **CSSO** inside an accredited **SCIF**	U.S. Citizen only; Tier 5 + CI/Full-Scope **Polygraph** + **Billet** + **Indoctrination** & Form 4414 **Special NDA**
AQ-SAP, IN-SAP, OS-SAP (Acknowledged & USAP / Black)	10 U.S.C. § 119(a)–(d); DoDD 5205.07; DoDM 5205.07; 32 CFR § 117.23 (SAP NISPOM)	DoD **SAPCO**, Component SAPCO, **PSO**, and **GSSO/CSSO** inside an accredited **SAPF**	**Tier 1** (program management) or **Tier 2** (operational/technical) SAP access; **Bigot List** & **PID** control; 30-day congressional notice for USAP
Waived SAP (WSAP / WUSAP / Deep Black) & Carve-Out	10 U.S.C. § 119(e), (g); 32 CFR § 117.6(b); DoDI 5205.11	SecDef / DepSecDef personal waiver; **SAPCO** & **PSO** exclusive oversight (DCSA relieved under **Carve-Out**)	**Tier 3** SAP access; oral/written notification restricted to 8 congressional defense leaders; **Enhanced Investigative** & CI Polygraph

1.3 Institutional Governance Agencies, Security Officers, and Billet Control

Oversight of national security enclaves is distributed across statutory Combat Support Agencies, Intelligence Community elements, and facility-level security officers:

- **National Security Agency (NSA):** Established under DoDD 5100.20, NSA serves as the signals intelligence (SIGINT) authority and the National Manager for National Security Systems, prescribing Type 1 cryptography, KMI, TEMPEST, and media sanitization standards.
- **Defense Intelligence Agency (DIA):** Established under DoDD 5105.21, DIA governs defense all-source military intelligence, operates the JWICS backbone, and serves as the DoD Cognizant Security Authority for military SCIFs (DoDM 5105.21 Vol 1).
- **National Reconnaissance Office (NRO):** Established under DoDD 5105.23, NRO designs, builds, launches, and operates the nation's overhead reconnaissance satellite constellations and exercises independent SAP/SCI and industrial Carve-Out authority.
- **National Geospatial-Intelligence Agency (NGA):** Established under DoDD 5105.60, NGA provides national geospatial intelligence (GEOINT) and manages GEOINT compartments across JWICS and SIPRNet.

- **Defense Information Systems Agency (DISA):** Established under DoDD 5105.19, DISA engineers and defends the Defense Information Systems Network (DISN), including NIPRNet, SIPRNet, DoD PKI, and the DoD Cloud Computing SRG boundary stack.
- **Defense Counterintelligence and Security Agency (DCSA) & Defense Security Service (DSS):** Reorganized under DoDD 5105.42 to succeed the legacy **DSS** and absorb OPM's National Background Investigations Bureau (NBIB), **DCSA** administers the National Industrial Security Program (NISP) and conducts over 95% of federal background investigations and Continuous Vetting.
- **Operational Security Officers (SAPCO, PSO, SSO, GSSO, CSSO, FSO, Security Officer):** The DoD **Special Access Program Central Office (SAPCO)** and Service/Component SAPCOs exercise central registry, billet ceiling, and policy oversight for all SAPs (DoDI 5205.11). At the program level, the **Program Security Officer (PSO)** manages overall **Program Security**, SCGs, PPPs, and SAPF operations. Within SCI compartments, a government **Special Security Officer (SSO)** or **Government Special Security Officer (GSSO)**—paired in cleared contractor facilities with a **Contractor Special Security Officer (CSSO)**—manages SCIF accreditation, DISS/Scattered Castles verification, and SCI indoctrinations (DoDM 5105.21 Vol 1). Within NISP industrial facilities, the contractor appoints a cleared U.S. citizen **Facility Security Officer (FSO)** (32 CFR § 117.7) who works alongside command **Security Officers** to enforce NISPOM compliance.

Access to any SCI or SAP compartment is strictly decouple-gated from general clearance eligibility. Even when an individual holds TS/SCI eligibility, access requires an affirmative **Need-to-Know (NTK)** determination and assignment to an authorized **Billet**—a specific funded personnel slot within an approved **Billet Plan** managed through a **Billet System** or SAPCO-controlled **Centralized Billet** structure (DoDM 5205.07; DoDM 5105.21 Vol 3). Every individual granted access to a sensitive compartment or SAP is recorded on an authoritative by-name roster historically and operationally termed a **Bigot List** (**Bigot**), which populates the physical facility **Access List** and the digital system **Access Control List (ACL)** keyed to a unique alpha-numeric or trigraph/digraph **Program Identifier (PID)**. Within SAPs, **Controlled Access** and **Tiered Access** divide exposure into **Tier 1** (programmatic/governance oversight), **Tier 2** (full technical/operational execution), and **Tier 3** (Waived SAP access) (DoDI 5205.11). Once approved, the candidate undergoes formal security **Indoctrination**—the **Read-In** / **Read-On** briefing—and executes both the standard Classified Information **Nondisclosure Agreement (NDA)** (SF-312) and a compartment-specific **Special NDA** (Form 4414 for SCI or the SAP Indoctrination NDA). Inside the enclave, logical and physical operations enforce **Role-Based Access Control (RBAC)** (NIST SP 800-53 AC-2), **Least Privilege** (AC-6), **Separation of Duties** (AC-5), physical/cryptographic **Dual Control**, **Two-Person Integrity (TPI)** (**Two-Person**), and physical **No-Lone-Zone (NLZ)** enforcement for Top Secret CRYPTO, nuclear, or Waived SAP assets. Upon billet termination or transfer, the individual must complete a formal administrative **Debrief** removing them from the Bigot List and ACL while reaffirming lifelong NDA obligations.

1.4 Personnel Vetting, Adjudication (SEAD 3/4/6/7), Polygraph, and Industrial/Export Controls

Under the Federal Investigative Standards (50 U.S.C. § 3341; E.O. 13467; DCSA Investigative Tiers), personnel vetting scales across four primary background investigation tiers: **Tier 1** (low-risk/HSPD-12 credentialing), **Tier 2** (moderate-risk public trust), **Tier 3** (Confidential and Secret national security eligibility), and **Tier 5** (Single Scope Background Investigation for Top Secret, SCI, and Q clearances), supplemented by **Enhanced Investigative** checks for SAP billets. Applicants submit the **Standard Form 86 (SF-86)** (Questionnaire for National Security Positions) through the legacy **e-QIP (Electronic Questionnaires for Investigations Processing)** portal or DCSA's modernized eApp/NBIS platform. Adjudicative and clearance records across the DoD and NISP have migrated from the retired **Joint Personnel Adjudication System (JPAS)** to the **Defense Information System for Security (DISS)** (comprising JVS and CATS) (DCSA DISS Architecture).

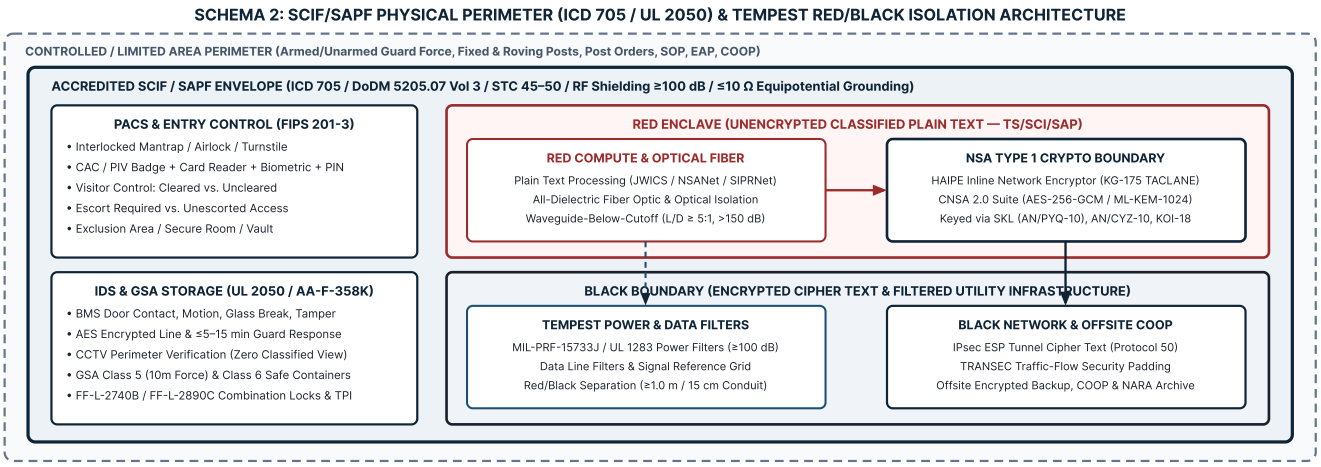
Instead of relying solely on periodic 5- or 10-year reinvestigations, Security Executive Agent Directive (SEAD) 6 mandates **Continuous Evaluation (CE)** and **Continuous Vetting (CV)** driven by continuous **Automated Records Checks (ARC)** across seven mandatory data categories (criminal, terrorism, financial, public records, foreign travel, credit, and eligibility databases). All background investigations and CE alerts undergo formal **Adjudicative** evaluation under SEAD 4 (National Security Adjudicative Guidelines) across 13 guidelines (Allegiance to the U.S., Foreign Influence, Foreign Preference, Sexual Behavior, Personal Conduct, Financial Considerations, Alcohol Consumption, Drug Involvement, Psychological Conditions, Criminal Conduct, Handling Protected Information, Outside Activities, and Use of Information Technology) to determine **Access Eligibility**. SEAD 3 imposes mandatory self- and peer-**Reporting** of unofficial **Foreign Travel**, continuing **Foreign Contact** with foreign nationals, foreign financial interests, and reportable arrest/financial events. SEAD 7 governs cross-agency reciprocity of background investigations and adjudications, complemented by mandatory **Annual Review**, **Access Review**, and **Eligibility Review** cycles (DoDM 5200.02).

For high-sensitivity IC, NSA, NRO, CIA, and SAP billets, vetting incorporates **Counterintelligence (CI)** screening (DoDD 5240.02) and physiological psychophysiological detection of deception—the **Polygraph** (DoDD 5210.48)—in three distinct scopes: a **Counterintelligence Polygraph (CI Poly)** testing espionage, sabotage, terrorism, unauthorized disclosure of classified data, and unreported foreign contacts; a **Lifestyle Polygraph (Lifestyle Poly / Expanded Scope Screening)** testing prior criminal conduct, falsification of security forms, and drug involvement; and a **Full Scope Polygraph (Full Scope Poly)** combining both CI and Lifestyle scopes. Personnel are strictly bifurcated into **Cleared** vs. **Uncleared** individuals, and by legal nationality status into a **U.S. Person (USP)** (E.O. 12333 § 3.5(k); 22 CFR § 120.62: U.S. citizens, lawful permanent residents, protected individuals under 8 U.S.C. § 1324b(a)(3), and U.S. corporations) versus a **Foreign National (FN)**. While LPRs qualify as USPs for unclassified export-controlled technical data, only U.S. citizens are eligible for TS/SCI/SAP clearances (ICD 704). Within the defense industrial base, the **National Industrial Security Program (NISP)** (E.O. 12829), codified in the **National Industrial Security Program Operating Manual (NISPOM)** at 32 CFR Part 117 and the **SAP NISPOM** supplement (32 CFR § 117.23 & DoDM 5205.07), enforces facility clearances, Foreign Ownership, Control, or Influence (FOCI) mitigation, and **Supply Chain** risk management. Simultaneously, statutory **Export Control** regimes—the Department of State's **International Traffic in Arms Regulations (ITAR)** (22 CFR Parts 120–130 governing the U.S. Munitions List) and the Department of Commerce's **Export Administration Regulations (EAR)** (15 CFR Parts 730–774 governing dual-use Commerce Control List items)—prohibit unauthorized physical or electronic transfer ("deemed export") of defense articles and technical data to any Foreign National.

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2. Physical Security Engineering (SCIF/SAPF), PACS/IDS, and Electromagnetic Security (TEMPEST/EMSEC)

Classified compartments and cryptographic roots of trust require a deterministic physical and electromagnetic envelope so that neither physical intrusion nor acoustic/radio-frequency emanations can compromise plaintext information. Under Intelligence Community Directive (ICD) 705, ICS 705-1, ICS 705-2, the IC Technical Specifications for Construction and Management of SCIFs (v1.5.1), and Unified Facilities Criteria UFC 4-010-05, every **Sensitive Compartmented Information Facility (SCIF)** and **Special Access Program Facility (SAPF)** (DoDM 5205.07) must complete formal **Facility Accreditation**—comprising **SCIF Accreditation** by the Cognizant Security Authority (DIA/ODNI/Service CSA), **SAPF Accreditation** by the SAPCO/AO, and **TEMPEST Accreditation** by a Certified TEMPEST Technical Authority (CTTA)—prior to introducing classified material.



Schema 2: ICD 705 / UL 2050 SCIF/SAPF Physical Security Perimeter, GSA Storage, and TEMPEST Red/Black Electromagnetic & Optical Isolation.

2.1 Hierarchical Security Zones, Vaults, GSA Containers, and Combination Locks

Military and industrial installations (DoDM 5200.08 Vol 3; DoDM 5200.01 Vol 3; 32 CFR Part 117) structure physical protection into concentric zones around the accredited **Facility**: an outer **Controlled Area** (buffer zone with administrative access checks), an intermediate **Limited Area** (where uncontrolled movement still does not permit direct access to classified material, defended by guards and barriers), and an inner **Exclusion Area** or **Secure Area** (where simple physical presence gives direct access to classified systems, synonymous with an Open Storage SCIF/SAPF, **Secure Room**, or **Vault**). Under IC Tech Spec v1.5.1 Chapters 3–4, a SCIF/SAPF perimeter requires true floor-slab-to-structural-ceiling construction, acoustic attenuation of **STC 45** (or **STC 50** where amplified audio operates), 16-gauge steel or 9-gauge expanded metal (ASTM F1267) / ¾-inch plywood structural backing for Open Storage walls outside a Controlled Area, and steel man-bars (½-inch bars on 6-inch centers) or acoustic/inspection baffles on all HVAC duct penetrations exceeding 96 square inches. A true **Vault** requires 8-inch reinforced concrete walls (#4 rebar on 6-inch grids) or GSA-approved modular vault panels (AA-V-2737) fitted with a Class 5 or Class 8 vault door (QPL-AA-D-600-12).

Inside Closed Storage facilities or for layered compartmentalization, classified documents, drives, and COMSEC keys are stored inside a **GSA Approved** security container (**Safe**) governed by Federal Specification AA-F-358K and FED-STD-809D (GSA Security Containers):

- **Class 5 Security Container:** Heavy steel armor cabinet rated for **10 man-minutes of covert/forced entry resistance**, **20 man-hours of surreptitious entry resistance**, **20 man-hours of lock manipulation resistance**, and **30 man-minutes of radiological attack resistance**; authorized for classified NSI, Top Secret CRYPTO, and Arms, Ammunition, and Explosives (AA&E).
- **Class 6 Security Container:** Unarmored filing cabinet rated for **20 man-hours of surreptitious entry resistance**, **20 man-hours of manipulation resistance**, and **20 man-hours of radiological resistance** (0 man-minutes forced entry); standard container for TS/SCI/SAP storage inside IDS-protected SCIFs/SAPFs.
- **Lock, Combination, and Key Control:** Every GSA container and SCIF/SAPF perimeter door requires an electromechanical **Lock** meeting Federal Specification FF-L-2740B (Kaba Mas X-10 or S&G 2740B) for containers and FF-L-2890C (CDX-10 pedestrian door deadbolt) for SCIF doors. Mechanical dial locks (FF-L-2937) are restricted strictly to non-NSI AA&E storage. Each **Combination** is classified at the highest level of the material stored inside, recorded on SF-700, and rotated upon personnel debrief or annually; physical maintenance keys are governed by strict administrative **Key Control** logs.

2.2 Physical Access Control Systems (PACS), Intrusion Detection (IDS), Guard Forces, and Continuity (COOP)

Entry into a SCIF or SAPF is enforced by an electronic **Physical Access Control System (PACS)** (**Access Control System / ACS** / **Access Control**) compliant with FIPS 201-3, NIST SP 800-116 Rev. 1, and IC Tech Spec v1.5.1 Chapter 8. All PACS decision controllers and databases must reside physically inside the SCIF/SAPF perimeter and communicate with exterior devices over encrypted OSDP v2 (AES-128) wiring. Entry requires multi-factor authentication combining a cryptographic identity **Badge** (CAC/PIV smart card verified via a contact/contactless **Card Reader** / **Reader**), a secret **Personal Identification Number (PIN)** keypad entry, and/or a physiological **Biometric** template (fingerprint or iris scanner enforcing a False Match Rate ≤ 0.001–0.0001). High-density perimeters enforce single-file passage and anti-tailgating through an optical/full-height **Turnstile** or an interlocked two-door **Mantrap** (**Airlock**) where door D_2 cannot unlatch until door D_1 is closed and locked (`NOT(Open(D1) AND Open(D2))`) and occupant weight/optical sensors verify a single person—or exact two-person presence where **Two-Person Integrity (TPI)** is enforced. Every **Visitor** is categorized into a **Cleared** visitor (whose clearance and indoctrination are verified in DISS/Scattered Castles, permitting **Unescorted** access only if read-in to that specific facility's compartments) versus an **Uncleared** or non-indoctrinated visitor, for whom **Escort Required** rules mandate continuous line-of-sight supervision by an indoctrinated **Escort**, sanitization/covering of all visible classified displays, and activation of overhead flashing warning beacons.

When unoccupied, the SCIF/SAPF is protected by an **Intrusion Detection System (IDS)** (**Intrusion Detection** / **Alarm**) certified under UL 2050 (National Industrial Security Systems) Extent 3 and IC Tech Spec v1.5.1 Chapter 9. Every perimeter door is monitored by a high-security triple-biased Balanced Magnetic Switch (BMS) **Door Contact** (UL 634 Level 2, triggering an alarm if gap displacement exceeds 3/8 inch). Interior volume is covered by 360-degree dual-technology **Motion** sensors (UL 639 combining Passive Infrared 8–14 µm thermal sensing and Microwave Doppler radar to detect four steps at one step per second) and acoustic/piezoelectric **Glass Break** sensors where applicable. All sensor housings, junction boxes, and Premise Control Units incorporate 24-hour active **Tamper** switches and NIST-validated AES-128/256 encrypted line supervision (UL 1610) backed by 24 hours of battery reserve. Closed-Circuit Television (**CCTV**) cameras monitor exterior portals and mantraps (positioned so lenses never view classified screens or keyboards). Any alarm or line-supervision fault triggers immediate **Response** by an **Armed** or **Unarmed** security **Guard Force** (**Guard** / **Force**) deployed across stationary **Fixed** entry **Post** assignments and mobile **Roving** patrols within the mandatory UL 2050 response window (≤ 5 minutes for proprietary/CRZH monitoring or ≤ 15 minutes for UL-listed central stations). Guard operations are bound by written **Post Orders** (**Orders**), facility **Standard Operating Procedures (SOP)**, and an **Emergency Action Plan (EAP)** governing fire, duress, or active-hostile evacuation without compromising classified material.

To guarantee mission survival during catastrophic disruptions, Federal Continuity Directive 1 (FCD 1), FCD 2, and NIST SP 800-34 Rev. 1 mandate a formal **Continuity of Operations (COOP)** (**Continuity**) and IT **Disaster Recovery** architecture capable of restoring Primary Mission Essential Functions at an alternate accredited SCIF/SAPF within 12 hours and sustaining operations for at least 30 days. Encrypted **Backup** datasets are replicated to geographically separated **Offsite** accredited **Secure Storage** vaults, while long-term historical records undergo formal NARA-compliant **Archive** and **Retention** management under 36 CFR Part 1234.

2.3 TEMPEST / EMSEC Engineering: Red/Black Separation, Shielding, Filtering, and Optical Isolation

Every electronic conductor carrying unencrypted baseband signals radiates electromagnetic fields and injects conducted transients onto power and ground networks. **Emission Security (EMSEC)**—codified under the **TEMPEST** program via CNSSI 4009, NSTISSAM TEMPEST/1-92, CNSSAM TEMPEST/01-13, CNSSI 7000,

and legacy NACSIM 5000—prevents adversary interception of **Compromising Emanations** (**Compromising** / **Emanations**). TEMPEST equipment certifications are stratified into Level I (1-meter inspectable space containment), Level II (20-meter containment), and Level III (100-meter containment).

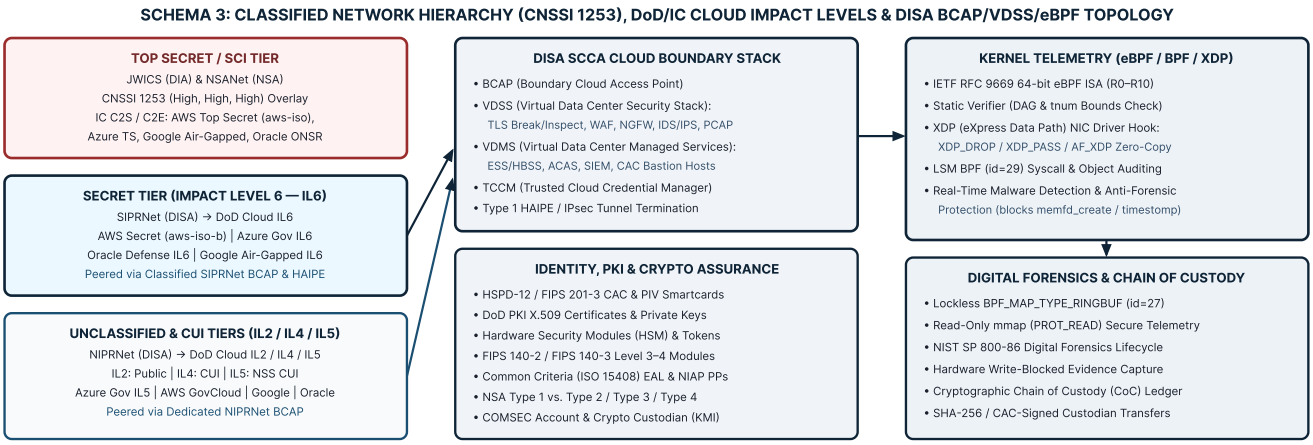
The foundational law of TEMPEST engineering is strict **Red/Black Separation** (**Red/Black**): **Red** designates any circuit, equipment, wireline, or **Red Network** that processes or carries unencrypted classified **Plain Text** (**Plain**), whereas **Black** designates any circuit, equipment, or **Black Network** that carries only NSA Type 1 encrypted **Cipher Text** (**Cipher**) or unclassified traffic. Only an NSA-certified Type 1 cryptographic module may bridge a Red-to-Black boundary. To prevent capacitive, inductive, or radiative cross-coupling from Red plain text onto Black lines (which exit the facility and act as antennas), CNSSAM TEMPEST/01-13 enforces physical separation distances (≥ 1.0 m between unshielded Red and Black equipment, ≥ 15 cm between metallic conduits, ≥ 2.5 cm for dielectric fiber, and mandatory 90° perpendicular crossings), supplemented by four electromagnetic barriers:

- **Electromagnetic Shielding:** Continuous 6-sided welded steel or modular copper/galvanized RF enclosures (MIL-HDBK-1195; IEEE Std 299 / MIL-STD-285) providing ≥ 100 dB magnetic, electric, and plane-wave attenuation across 150 kHz to 10 GHz.
- **Power and Data Line Filtering:** Every AC/DC **Power Line** (**Power**) and metallic **Data Line** (**Data**) penetrating the shielded SCIF/SAPF boundary must pass through a bulkhead-mounted **Power Line Filter** (**Power Filtering** / **Filtering**) or **Data Line Filter** certified under MIL-PRF-15733J / UL 1283 to deliver ≥ 100 dB insertion loss from 14 kHz to 10 GHz, blocking conducted RF leakage back onto the commercial utility grid.
- **Equipotential Grounding:** Under MIL-STD-188-124B and MIL-HDBK-419A, facilities implement a ≤ 10 Ω (≤ 5 Ω preferred) Earth Electrode Subsystem bonded to a high-frequency copper Signal Reference Grid with Class A RF bonds (≤ 1.0 mΩ DC resistance) and strict single-point grounding between Red and Black chassis to eliminate ground-loop currents.
- **Fiber Optic Transmission and Optical Isolation:** Because all-dielectric glass **Fiber Optic** (**Fiber** / **Optical**) cables (lacking metallic strength members or armor) neither radiate RF energy nor conduct ground currents, classified Red networks rely on optical media and **Optical Isolation** (opto-isolators and Waveguide-Below-Cutoff penetration tubes with length-to-diameter ratio L/D ≥ 5:1, providing > 150 dB evanescent RF attenuation below the waveguide cutoff frequency `f_c = 175.8/D_{\text{mm}\text{GHz}}`).

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3. Classified Network Topology (CNSSI 1253), DoD/IC Cloud Enclaves, PKI/HSM Identity, and Type 1 Cryptographic Infrastructure

Unlike civilian federal information systems that collapse security categorization into a single FIPS 199 scalar, National Security Systems are governed by CNSSI 1253 (Security Categorization and Control Selection for National Security Systems). CNSSI 1253 evaluates Confidentiality, Integrity, and Availability as an independent three-tuple $SC_{NSS} = \{(C, v_C), (I, v_I), (A, v_A)\}$ across Low, Moderate, and High tiers ($3^3 = 27$ permutations) mapped directly to NIST SP 800-53 Rev. 5 controls, augmented by Classified Information, Space, Cross-Domain Solution, and Intelligence Community overlays.



Schema 3: CNSSI 1253 Classified Networks (JWICS/NSANet/SIPRNet/NIPRNet), DoD/IC Cloud Enclaves (IL2–IL6 & C2S), DISA SCCA Boundary Stack, and Kernel eBPF/XDP Telemetry.

3.1 Sovereign Classified Networks and Enclave Boundaries

Each sovereign **Classified Network** (**Network**) operates as an isolated routing and cryptographic domain linking accredited **Classified Enclave** and **Secure Enclave** boundaries (CNSSI 4009):

- **Joint Worldwide Intelligence Communications System (JWICS):** Operated by **DIA** (DIA DoDIIS / JWICS Architecture), JWICS is the global IPv4/IPv6 MPLS network transporting **Top Secret / SCI (TS/SCI)** data, video, and sensor feeds between ICD 705 SCIFs worldwide, categorized under CNSSI 1253 as (High, High, Moderate/High) and encrypted end-to-end over the Black DISN optical core via NSA Type 1 HAIPE inline network encryptors.
- **NSANet:** Operated by **NSA** (NSA Enterprise Network), NSANet is the sovereign cryptologic SIGINT/cyber intranet operating at `TOP SECRET // SI / TK` (High, High, High), segregated from JWICS by hardware Cross-Domain Solutions (CDS), independent BGP autonomous systems, and continuous TRANSEC traffic-flow padding.
- **Secret Internet Protocol Router Network (SIPRNet / SIPR):** Operated by **DISA** (DISN Connection Process Guide), SIPRNet is the Department of Defense's primary operational command-and-control network for Collateral **Secret** data (up to SECRET//REL), using split-horizon `.smil.mil` / `.sgov.gov` DNS, NSS PKI hardware tokens, and either Protected Distribution Systems (CNSSI 7003) or Type 1 HAIPE WAN encryption.
- **Non-classified Internet Protocol Router Network (NIPRNet / NIPR):** Operated by **DISA** (DoD PKI / NIPRNet), NIPRNet transports Unclassified and Controlled Unclassified Information (CUI) (Low/Moderate, Moderate, Low/Moderate), enforces perimeter inspection at DISA Internet Access Points (IAPs) and Joint Regional Security Stacks (JRSS), and serves as the underlying "Black Core" transport carrying outer HAIPE IPsec ESP packets for SIPRNet and JWICS.

3.2 DoD & IC Cloud Enclaves (IL2, IL4, IL5, IL6, C2S) and DISA SCCA Boundary Stack (BCAP, VDSS, VDMS, TCCM)

Under the DISA Cloud Computing Security Requirements Guide (CC SRG) and the DoD CIO Cloud Security Playbook, every **DoD Cloud**, **Secure Cloud**, or **Commercial Cloud** environment operated by hyper-scale Cloud Service Providers—specifically Amazon Web Services (**AWS Secret**, **AWS Top Secret**), Microsoft (**Azure Government**, **Azure IL5**, **Azure IL6**), **Google** (Google Cloud Assured Workloads & Google Distributed Cloud Air-Gapped), and **Oracle** (Oracle Cloud Infrastructure Defense & National Security Regions)—is governed by rigorous data classification **Impact Levels (Impact Level)**:

- **Impact Level 2 (IL2):** Non-Controlled Unclassified Information approved for public release (FedRAMP Moderate baseline; internet-accessible via DISA IAPs).
- **Impact Level 4 (IL4):** Controlled Unclassified Information (CUI) not designated as National Security Systems data; requires private circuit peering via a NIPRNet **Boundary Cloud Access Point (BCAP)**.
- **Impact Level 5 (IL5):** Higher-sensitivity CUI, mission-critical data, and Unclassified National Security Systems (NSS) data; mandates U.S. Citizen / U.S. soil sovereign administration, FIPS 140-3 Level 3 HSM Customer-Managed Encryption Keys, and dedicated NIPRNet BCAP transport (e.g., **Azure Government** / **Azure IL5**, AWS GovCloud, **Google** Assured Workloads IL5, **Oracle** DoD IL5).
- **Impact Level 6 (IL6):** Classified Collateral **Secret** information (up to `SECRET//NOFORN`); mandates physically air-gapped facilities accredited to Secret open-storage standards, peered directly to **SIPRNet** via NSA Type 1 HAIPE encryptors and a classified SIPRNet BCAP (e.g., **AWS Secret** `aws-iso-b`, **Azure IL6** / Azure Government Secret, **Oracle** Defense Cloud IL6, **Google** Air-Gapped IL6).
- **Commercial Cloud Services (C2S) & C2E (Top Secret / SCI Cloud):** Pioneered by the CIA and expanded across the Intelligence Community and DoD JWCC, **C2S** (and its multi-vendor successor C2E) operates inside ICD 705 SCIF data centers peered directly to **JWICS** and **NSANet**, hosting **AWS Top Secret** (`aws-iso` and `aws-iso-f`), Azure Top Secret, Google Air-Gapped TS, and Oracle National Security Regions.

To prevent direct uninspected routing between a commercial cloud VPC/VNet and the DISN backbone, the DISA Secure Cloud Computing Architecture (SCCA) mandates four integrated boundary components (DoD Cloud Playbook; AWS DoD SCCA Reference; Microsoft SACA): (1) the **Boundary Cloud Access Point (BCAP)**, DISA's physical demarcation point terminating CSP dedicated optical circuits (Direct Connect, ExpressRoute, Cloud Interconnect, FastConnect), enforcing BGP route filtering, terminating Type 1 HAIPE tunnels at IL6, and mirroring packets to DISA Cyber Security Assurance Analytics Cloud (CSAAC) sensors; (2) the **Virtual Data Center Security Stack (VDSS)**, a virtualized perimeter inspection hub inside the cloud enclave chaining TLS 1.3 break-and-inspect proxies, Web Application Firewalls, Layer 3–7 Next-Generation Firewalls, IDS/IPS, and full packet capture; (3) the **Virtual Data Center Managed Services (VDMS)**, an out-of-band management enclave hosting DoD Endpoint Security Solutions (ESS/HBSS), Assured Compliance Assessment Solution (ACAS) vulnerability scanners, SIEM log forwarders, and CAC-enforced bastion jump hosts; and (4) the **Trusted Cloud Credential Manager (TCCM)** (also referencing forward-deployed **Theater/Tactical Cloud Computing** edge appliances in DDIL environments), a dedicated governance role and hardware-credentialed bastion controlling root cloud tenant accounts and IAM break-glass keys.

3.3 Hardware Identity (HSPD-12, FIPS 201-3, CAC/PIV, PKI, HSM) and Cryptographic Assurance (FIPS 140-3, Type 1, HAIPE, KMI)

Zero-trust identity across NIPRNet, SIPRNet, and cloud enclaves is anchored in Homeland Security Presidential Directive 12 (HSPD-12), FIPS 201-3 (Personal Identity Verification of Federal Employees and Contractors) (**FIPS 201**), and DoDI 8520.02. Every user authenticates using a hardware **Personal Identity Verification (PIV)** smart card or DoD **Common Access Card (CAC)** integrated into the hierarchical DoD **Public Key Infrastructure (PKI)** (**CAC/PKI**). Each CAC/PIV smart card is a tamper-resistant cryptographic **Token** containing a validated **Crypto Module** that generates and stores non-exportable asymmetric **Private Key** pairs (`CKA_EXTRACTABLE = FALSE`) bound to X.509v3 digital **Certificate** chains across four standardized NIST SP 800-73-4 slots: Slot 9A (PIV/CAC network authentication), Slot 9C (non-repudiation digital signature requiring per-operation PIN entry), Slot 9D (key management / S/MIME decryption, escrowed in the DoD Key Recovery Authority), and Slot 9E (card authentication for physical PACS readers). At the infrastructure tier, Certificate Authority root keys and cloud Customer-Managed Encryption Keys are safeguarded inside a network or PCIe **Hardware Security Module (HSM)** (NIST CSRC HSM).

Cryptographic modules and cybersecurity products undergo dual validation pathways: NIST's Cryptographic Module Validation Program evaluates every hardware, software, or firmware **Crypto Module** against FIPS 140-3 (and historical FIPS 140-2, collectively **FIPS 140**) across Security Levels 1 through 4 (where Level 3 requires tamper-evident coatings and identity-based auth, and Level 4 mandates active environmental fault envelopes and immediate battery-backed zeroization upon physical penetration). Concurrently, broader IT products are evaluated under the international Common Criteria (ISO/IEC 15408) framework across **Evaluation Assurance Levels (EAL)** EAL 1 through EAL 7 and NIAP Protection Profiles. For National Security Systems, the NSA classifies cryptographic equipment into four product assurance tiers: **Type 1** (NSA-endorsed Controlled Cryptographic Items approved for classified TS/SCI/SAP and Secret NSI, incorporating suite-grade hardware tamper protection and red/black failure isolation), **Type 2** (NSA-endorsed equipment for unclassified NSS), **Type 3** (NIST FIPS-validated commercial algorithms for CUI/SBU), and **Type 4** (unevaluated commercial or proprietary algorithms). Under the Commercial National Security Algorithm Suite 2.0 (CNSA 2.0), NSA mandates quantum-resistant cryptography across NSS: `AES-256` symmetric encryption, `SHA-384/SHA-512` hashing, `ML-KEM-1024` (FIPS 203 CRYSTALS-Kyber) key establishment, `ML-DSA-87` (FIPS 204 CRYSTALS-Dilithium) digital signatures, and stateful hash-based `LMS/XMSS` firmware signatures.

Within the broader discipline of **Information Security (INFOSEC)** and **Operations Security (OPSEC)**, telecommunications protection is governed by **Communications Security (COMSEC)** (DoDI 8523.01) and **Transmission Security (TRANSEC)** (which conceals traffic patterns via frequency hopping, spread spectrum, and deterministic packet-length/timing padding). Across wide-area networks, Type 1 **Encryption** (**Crypto**) is performed by a **High Assurance Inline Encryptor** / **Network Encryptor** (**Encryptor**) compliant with the NSA **High Assurance Internet Protocol Encryptor (HAIPE)** Interoperability Specification (v4.x). The flagship HAIPE hardware family is the General Dynamics Mission Systems KG-175 TACLANE series (**KG-175** / **TACLANE**, including the KG-175D Micro, KG-175G 2 Gbps, KG-175X 20 Gbps, and TACLANE-E100 200 Gbps). A KG-175 TACLANE sits physically across the SCIF Red/Black boundary, ingesting unencrypted Red IPv4/IPv6 packets, encrypting the entire inner IP datagram with AES-256-GCM inside an outer Black **IPsec** Encapsulating Security Payload (ESP, IP Protocol 50) tunnel header, and applying TRANSEC traffic-flow security padding.

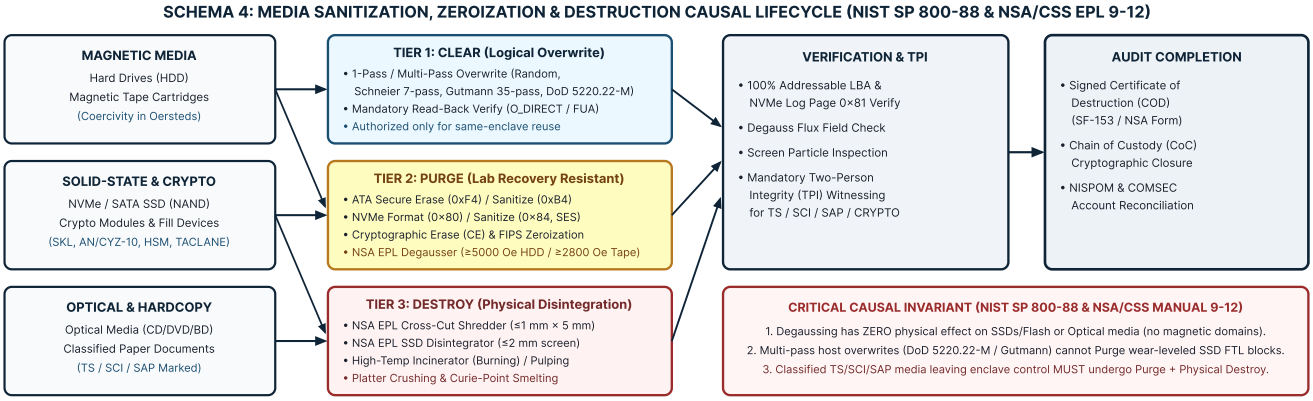
All cryptographic **Key Management** and **Keying Material (KEYMAT)** distribution is tracked inside NSA's Electronic Key Management System / Key Management Infrastructure (EKMS / KMI). Every organization holding physical key material or Controlled Cryptographic Items operates a registered **COMSEC Account** supervised by a formally appointed **Crypto Custodian** (COMSEC Account Manager / KOA Manager) subject to 100% physical inventory audits and SF-153 accounting. Physical injection of KEYMAT into a KG-175 TACLANE or tactical radio is performed via an electronic **Fill Device** (Army TM 11-5810-292-13&P) over a 6-pin `DS-102` (biphase serial) or `DS-101` (packetized RS-485) fill port: the legacy optical paper-tape reader KOI-18 (which pulls punched Mylar/paper key tape past photodiodes with zero internal memory), the EKMS-era portable Data Transfer Device AN/CYZ-10, and the modern KMI-integrated Simple Key Loader (SKL, AN/PYQ-10 / AN/PYQ-10C) housing an embedded Type 1 KOV-21 cryptographic engine.

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4. Kernel-Level Telemetry (eBPF/XDP), Digital Forensics, and Media Sanitization & Destruction Lifecycle

Inside classified cloud and SCIF compute nodes, host and network assurance relies on deterministic kernel-space observability to provide real-time **Secure Telemetry**, **Malware Detection**, and defense against **Anti-Forensic** evasion without introducing unstable third-party kernel modules. Modern Linux-based enclave hosts implement the **Extended Berkeley Packet Filter (eBPF)** and classic **BPF** architecture standardized in IETF RFC 9669 (BPF Instruction Set Architecture) and Linux Kernel eBPF Documentation. Unlike classic 32-

bit cBPF (two registers A and X), eBPF defines a 64-bit RISC virtual machine with eleven 64-bit registers (`R0`–`R10`, where `R10` is a read-only frame pointer to a 512-byte stack) and 64-bit instructions (plus 128-bit `BPF_LD_IMM64 0x18`). Prior to JIT compilation into native `x86_64` or `arm64` machine code, every eBPF program must pass the in-kernel static verifier, which performs directed acyclic graph (DAG) / bounded-loop control-flow verification (up to 1,000,000 verified instructions) and abstract register-state tracking (`struct bpf_reg_state` and tristate `tnum` masks) to prove zero out-of-bounds memory dereferences, zero uninitialized reads, and guaranteed termination.



Schema 4: NIST SP 800-88 Rev. 1 / Rev. 2 and NSA/CSS Policy Manual 9-12 Media Sanitization, Cryptographic Zeroization, and Physical Destruction Lifecycle.

4.1 XDP Line-Rate Packet Inspection, LSM BPF Anti-Forensics, and Digital Forensics Chain of Custody

At the network ingress boundary of an enclave host, the **eXpress Data Path (XDP)** (Linux Kernel AF_XDP & XDP) executes verified eBPF programs directly inside the network interface card (NIC) driver's NAPI receive poll loop—or offloaded directly onto SmartNIC silicon (`XDP_FLAGS_HW_MODE`)—before the kernel allocates an `sk_buff` structure. Operating on raw DMA packet frames (`struct xdp_buff`), XDP enforces deterministic verdicts at line rate: `XDP_DROP (1)` to discard malformed or unauthorized cross-domain packets, `XDP_TX (3)` to bounce traffic, `XDP_REDIRECT (4)` for zero-copy frame steering into `AF_XDP` (`XSKMAP`) userspace inspection rings, or `XDP_PASS (2)` to admit validated traffic to the network stack.

Above the network driver, eBPF attaches to kernel tracepoints, kprobes, and **Linux Security Module (LSM)** hooks (`BPF_PROG_TYPE_LSM`, `id = 29`; Linux LSM BPF) to enforce mandatory access control and defeat **Anti-Forensic** techniques. Because LSM BPF hooks inspect resolved kernel objects (`struct linux_binprm`, `struct inode`, `struct file`) rather than raw userspace string pointers, they eliminate Time-of-Check to Time-of-Use (TOCTOU) race conditions, blocking fileless malware execution via anonymous memory descriptors (`memfd_create + execveat`), unauthorized `ptrace` injection, log tampering, or timestamp manipulation (`utimensat`). Audit events are streamed to userspace collectors via lockless single-producer/single-consumer ring buffers (`BPF_MAP_TYPE_RINGBUF`, `id = 27`; BPF Ringbuf) mapped read-only (`PROT_READ`) to userspace so compromised processes cannot overwrite historical telemetry. When an anomaly is detected, incident responders execute the four-phase **Digital Forensics** (**Forensics**) lifecycle defined in NIST SP 800-86 (Guide to Integrating Forensic Techniques into Incident Response): Collection (capturing artifacts strictly in Order of Volatility from CPU registers and RAM to non-volatile storage using hardware write-blockers), Examination, Analysis, and Reporting. Admissibility and audit integrity of all captured **Evidence** rely on an unbroken cryptographic and administrative **Chain of Custody (CoC)**, logging every physical/digital transfer along with SHA-256/SHA-512 digests (H_0 = SHA-256(E)) signed by hardware PIV/CAC credentials.

4.2 Media Sanitization, Cryptographic Erase, Zeroization, and Physical Destruction (NIST SP 800-88 & NSA/CSS Manual 9-12)

When classified or sensitive storage **Media**—spanning magnetic **Hard Drive** (HDD) platters, magnetic **Tape** cartridges, NAND flash **Solid-State Drive (SSD)** assemblies, **Optical** discs (CD/DVD/Blu-ray), and hardcopy **Paper Document** records—reaches end-of-life, reclassification, or decommissioning, simple administrative **Disposal** is strictly prohibited. Instead, NIST SP 800-88 Rev. 1 (and its September 2025 update NIST SP 800-88 Rev. 2), 32 CFR § 117.15(e)(2) (NISPOM), and NSA/CSS Policy Manual 9-12 (Storage Device Sanitization and Destruction Manual) mandate a deterministic **Media Sanitization** (**Sanitization**) and **Secure Disposal** workflow across three escalating technical tiers:

1. **Clear (Logical Overwrite):** Applies programmatic software or firmware **Overwrite** across all user-addressable Logical Block Addresses (LBAs) using fixed patterns (`0x00`, `0xFF`) or cryptographically strong pseudorandom bitstreams (**Random**) with Direct I/O (`O_DIRECT`) and Force Unit Access (FUA) to bypass OS caches, followed by read-back **Verify** checking. Historically, multi-pass overwrite schedules dominated magnetic media sanitization: the 1995 **DoD 5220.22-M** 3-pass protocol (Pass 1: character, Pass 2: bitwise complement, Pass 3: random + verify), Bruce **Schneier**'s 7-pass protocol (Applied Cryptography, 1996: `0xFF`, `0x00`, and five CSPRNG passes), and Peter **Gutmann**'s 35-pass schedule (USENIX Security 1996: 4 random passes, 27 deterministic MFM/RLL 3-byte flux patterns `0x555555`–`0xDB6DB6`, and 4 random passes). Under modern 32 CFR Part 117 and NIST SP 800-88, DoD 5220.22-M multi-pass recipes are formally superseded: a single random/zero pass with read-back verification suffices to *Clear* modern PRML/EPRML HDDs for reuse within the same security enclave, whereas host-level overwrites are strictly insufficient to *Purge* wear-leveled SSDs due to unaddressable Flash Translation Layer (FTL) over-provisioning blocks.
2. **Purge (Laboratory-Resistant Sanitization):** Renders target data infeasible to recover even using state-of-the-art laboratory attacks (Magnetic Force Microscopy or NAND chip-off analysis). Purge is executed via three primary mechanisms:
   - *Controller Firmware Sanitize & Secure Erase Commands:* On SATA/ATA drives (INCITS 529 ACS-4), issuing **ATA Secure Erase** (**Secure Erase**: `SECURITY ERASE PREPARE` `0xF3` followed by `SECURITY ERASE UNIT` `0xF4` with Enhanced bit 1) or the modern `SANITIZE DEVICE` (`0xB4`) opcode (`0x0012` Block Erase `"BkEr"`, `0x0014` Overwrite `"OWrt"`, or `0x0011` Crypto Scramble `"Cryp"`) purges active LBAs, remapped bad sectors (`G-List`), and hidden HPA/DCO areas. On PCIe/NVMe SSDs (NVM Express Base Specification), issuing **NVMe Format** (`Format NVM` Admin Opcode `0x80` with `CDW10.SES = 001b` User Data Erase or `010b` Cryptographic Erase) or the subsystem-wide `Sanitize` command (Admin Opcode `0x84`, `SANACT = 010b` Block Erase or `100b` Crypto Erase, verified via Telemetry Log Page `0x81`) electrically drains charge from all floating-gate/3D TLC/QLC charge-trap NAND blocks across the entire NVM subsystem.
   - *Cryptographic Erase (CE), Key Destruction, and Zeroization:* On self-encrypting drives (SEDs) and FIPS 140-3 cryptographic modules (HSMs, KG-175 TACLANEs, SKLs), **Cryptographic Erase (CE)** performs high-assurance **Key Destruction** and active hardware **Zeroization** (FIPS 140-3 IG § 9.3) of all plaintext Media Encryption Keys (`MEK`) and Key Encryption Keys (`KEK`) across volatile SRAM and non-volatile EEPROM/flash, leaving only AES-XTS-256 ciphertext on the physical media.
   - *Magnetic Degaussing:* For magnetic **Hard Drive** platters and magnetic **Tape** only, passing the media through an **NSA Approved** capacitive-discharge or rare-earth permanent-magnet **Degausser** listed on the NSA/CSS Evaluated Products List (EPL) (generating peak magnetic flux density ≥ 1.0–2.0 Tesla / ≥ 5,000 Oersteds for perpendicular HDD platters and ≥ 2,800 Oe for magnetic tape) permanently randomizes magnetic domain alignments and destroys servo factory tracks. Because SSDs and Optical

discs store data via trapped charge or physical polycarbonate pits rather than magnetic domains, degaussing has zero effect on SSD or Optical media.

3. **Destroy (Physical Disintegration, Shredding, Burning, and Pulping):** Under NSA/CSS Policy Manual 9-12, all classified TS/SCI/SAP media leaving physical enclave control must undergo physical **Destruction** (**Destroy**) using equipment on the **NSA Approved** Evaluated Products List:
   - *Classified Paper & Flexible Mylar/Tape:* High-security **Cross-Cut Shredding** in an NSA EPL **Shredder** reducing material to a maximum particle size of **≤ 1 mm × 5 mm** (≤ 5 mm^2 area, DIN 66399 Level P-7), wet chemical/mechanical **Pulping** through a ≤ 1/4-inch (6.35 mm) screen, or complete thermal **Burning** inside an approved high-temperature **Incinerator** until reduced to white ash.
   - *Solid-State Drives (SSD) & Optical Media:* Mechanical disintegration in an NSA/CSS EPL Solid State Disintegrator equipped with a **≤ 2 mm nominal screen** (shearing sub-centimeter BGA NAND dies into ≤ 2 mm fragments) or surface milling/incineration for **Optical** discs.
   - *Magnetic Hard Drives (HDD):* Degaussing in an NSA EPL Degausser followed by hydraulic platter crushing/bending or rotary disintegration, or thermal smelting above the alloy Curie temperature (T_C > 750°C–1,115°C).

Every sanitization and destruction action on classified or COMSEC media must be executed and witnessed under **Two-Person Integrity (TPI)** and formally recorded on a signed **Certificate of Destruction (COD)** (e.g., SF-153 for COMSEC or NSA/DoD media destruction certificates; NIST SP 800-88 Rev. 1 Appendix G), permanently closing the asset's lifecycle in the **Chain of Custody (CoC)** and NISPOM/COMSEC inventory ledgers.

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5. Consolidated Master Audit Matrix of All 265 Taxonomy Terms (9.txt)

The tables below provide the complete, auditor-verifiable reference mapping all 265 unique terms from `9.txt` to their governing statutory/engineering standard, functional architecture, and deterministic causal role.

5.1 Governance, Classification, SAP/SCI, Billet Control, Vetting, and Export Control Matrix

Taxonomy Terms (Exact Match)	Primary Governing Authority	Engineering, Administrative & Causal Specification
ORIGINAL CLASSIFICATION AUTHORITY, OCA, DERIVATIVE, CLASSIFIER, CLASSIFICATION, CLASSIFIED, COLLATERAL, TS, S, C	E.O. 13526 §§ 1.1–2.1; 32 CFR Part 2001; DoDM 5200.01 Vol 1	**OCA** executes initial national security damage determination assigning **Classified** tiers: **TS** (Top Secret, exceptionally grave damage), **S** (Secret, serious damage), or **C** (Confidential, damage). Non-compartmented clearances constitute **Collateral** security. A **Derivative Classifier** (**Derivative** / **Classifier**) propagates classification from an SCG or source document.
MARKING, BANNER, HEADER, FOOTER, PORTION, PORTION MARKING, DECLASS, DECLASSIFICATION, DOWNGRADE, DOWNGRADING, UPGRADE, RECLASSIFY	32 CFR §§ 2001.21–2001.30; DoDM 5200.01 Vol 2; ICD 703	Mandatory **Marking** places overall classification **Banner** lines in the page **Header** and **Footer** and parenthetical **Portion Marking** (**Portion**) on every paragraph/table/figure. Lifecycle transitions comprise **Declassification** (**Declass**), **Downgrading** (**Downgrade**), **Upgrading** (**Upgrade**), and agency-head **Reclassifying** (**Reclassify**).

Taxonomy Terms (Exact Match)	Primary Governing Authority	Engineering, Administrative & Causal Specification
SECURITY CLASSIFICATION GUIDE, SCG, PROGRAM PROTECTION, PROGRAM PROTECTION PLAN, PPP, CRITICAL PROGRAM INFORMATION, CPI, SYSTEM SECURITY PLAN, SECURITY PLAN, SSP	DoDM 5200.45; DoDI 5200.39; DoDI 5000.83; NIST SP 800-18 Rev. 2	The OCA-signed **SCG** codifies element-level classification rules; the acquisition **PPP** (**Program Protection**) safeguards **CPI** via anti-tamper and SCRM engineering; the information system **SSP** (**Security Plan**) documents all implemented NIST SP 800-53 / CNSSI 1253 security controls for ATO authorization.
RISK ASSESSMENT, THREAT, VULNERABILITY, IMPACT, MITIGATION, CONTROL, SAFEGUARD, COUNTERMEASURE, COUNTERMEASURES, TECHNICAL, ADMINISTRATIVE, PHYSICAL, PERSONNEL, INDUSTRIAL	NIST SP 800-30 Rev. 1; NIST SP 800-53 Rev. 5; 32 CFR Part 117	**Risk Assessment** quantifies risk from an adversary **Threat** exploiting a **Vulnerability** to cause mission **Impact**, applying risk **Mitigation** via a layered **Control**, **Safeguard**, and **Countermeasure** (**Countermeasures**) baseline across **Technical**, **Administrative**, **Physical**, **Personnel**, and **Industrial** domains.
COMPARTMENTED, SCI, TS/SCI, UMBRELLA, COMPARTMENT, SUB-COMPARTMENT, ENHANCED SAFEGUARDING	E.O. 13526 § 4.3; ICD 704; DoDM 5105.21 Vols 1–3	**Compartmented** intelligence regimes (**SCI** / **TS/SCI**) apply **Enhanced Safeguarding** organized under an **Umbrella** control system (e.g., SI, TK), divided into a functional **Compartment** and granular **Sub-Compartment** to isolate sensitive intelligence sources and methods.
SAP, AQ-SAP, IN-SAP, OS-SAP, STRUCTURAL SAP, CONTENT SAP, ACKNOWLEDGED SAP, UNACKNOWLEDGED, USAP, BLACK, BLACK PROGRAM, WAIVED, WSAP, WUSAP, DEEP BLACK, CARVE-OUT	10 U.S.C. § 119; DoDD 5205.07; DoDI 5205.11; 32 CFR § 117.6(b)	**SAP** categories span Acquisition (**AQ-SAP**), Intelligence (**IN-SAP**), and Operations/Support (**OS-SAP**), structured as a **Structural SAP** or **Content SAP**. Disclosure tiers progress from **Acknowledged SAP** to **Unacknowledged** (**USAP** / **Black** / **Black Program**, § 119(d)) and SecDef-**Waived** (**WSAP** / **WUSAP** / **Deep Black**, § 119(e),(g) 8-leader reporting). **Carve-Out** exempts SAP contracts from routine DCSA inspection.
NSA, DIA, NRO, NGA, DISA, DSS, DCSA, SAPCO, PSO, PROGRAM SECURITY OFFICER, PROGRAM SECURITY, SSO, GSSO, CSSO, FSO, SECURITY OFFICER	DoDD 5100.20; 5105.21; 5105.23; 5105.60; 5105.19; 5105.42; 32 CFR § 117.7	Institutional authorities **NSA**, **DIA**, **NRO**, **NGA**, **DISA**, and **DCSA** (succeeding **DSS**) oversee national security enclaves; **SAPCO** and the **PSO** (**Program Security Officer**) direct **Program Security** for SAPs; **SSO**, **GSSO**, and contractor **CSSO** manage SCI/SCIFs; contractor **FSO** and command **Security Officer** enforce collateral/NISP compliance.
BILLET, BILLET PLAN, BILLET SYSTEM, CENTRALIZED BILLET, BIGOT, BIGOT LIST, ACCESS LIST, ACCESS CONTROL LIST, PROGRAM IDENTIFIER, CONTROLLED ACCESS,	DoDM 5205.07; DoDI 5205.11; DoDM 5105.21 Vol 3	Authorized personnel slots (**Billet** / **Billet Plan** / **Billet System** / **Centralized Billet**) cap compartment population. Indoctrinated personnel are tracked on a **Bigot List** (**Bigot**), physical **Access List**, and digital **Access Control List** bound to a **Program Identifier** under **Controlled Access** and **Tiered Access** (**Tier 1**, **Tier 2**, **Tier 3**).

Taxonomy Terms (Exact Match)	Primary Governing Authority	Engineering, Administrative & Causal Specification
TIERED ACCESS, TIER 1, TIER 2, TIER 3		
NEED-TO-KNOW, NTK, ROLE-BASED ACCESS, RBAC, LEAST PRIVILEGE, SEPARATION OF DUTIES, DUAL CONTROL, TWO-PERSON INTEGRITY, TWO-PERSON, TPI, NO-LONE-ZONE, NLZ, READ-IN, READ-ON, INDOCTRINATION, DEBRIEF, NDA, NONDISCLOSURE AGREEMENT, SPECIAL NDA	E.O. 13526 § 4.1; NIST SP 800-53 Rev. 5 (AC-2, AC-5, AC-6); DoDM 5205.07 Vol 4	Access requires verified **Need-to-Know (NTK)**, formal **Indoctrination** (**Read-In** / **Read-On**), execution of the SF-312 **Nondisclosure Agreement (NDA)** and Form 4414/SAP **Special NDA**, and operational enforcement of **Role-Based Access (RBAC)**, **Least Privilege**, **Separation of Duties**, **Dual Control**, **Two-Person Integrity (TPI / Two-Person)**, and **No-Lone-Zone (NLZ)** rules, concluding with a formal **Debrief**.
TIER 5, SF-86, eQIP, JPAS, DISS, CONTINUOUS VETTING, CV, CONTINUOUS EVALUATION, CE, AUTOMATED RECORDS CHECK, ARC, SEAD, SEAD 3, SEAD 4, SEAD 6, SEAD 7, ENHANCED INVESTIGATIVE, ADJUDICATIVE, ACCESS ELIGIBILITY, ANNUAL REVIEW, ACCESS REVIEW, ELIGIBILITY REVIEW	50 U.S.C. § 3341; SEAD 3, SEAD 4, SEAD 6, SEAD 7; DoDM 5200.02	Vetting progresses across **Tier 1**, **Tier 2**, **Tier 3**, and **Tier 5** (plus SAP **Enhanced Investigative** checks) initiated via **SF-86** on **eQIP** / NBIS and recorded in **DISS** (succeeding **JPAS**). **SEAD 6** mandates **Continuous Vetting (CV)** and **Continuous Evaluation (CE)** using **Automated Records Checks (ARC)**; **SEAD 4** governs 13-guideline **Adjudicative** determinations of **Access Eligibility**; **SEAD 7** enforces reciprocity alongside **Annual Review**, **Access Review**, and **Eligibility Review** cycles.
POLYGRAPH, CI POLY, LIFESTYLE POLY, FULL SCOPE POLY, COUNTERINTELLIGENCE, CI, FOREIGN CONTACT, FOREIGN TRAVEL, REPORTING, CLEARED, UNCLEARED, US PERSON, USP, FOREIGN NATIONAL, FN, NISP, NISPOM, SAP NISPOM, ITAR, EAR, EXPORT CONTROL, SUPPLY CHAIN	DoDD 5210.48; DoDD 5240.02; 32 CFR Part 117; 22 CFR Parts 120–130; 15 CFR Parts 730–774	**Counterintelligence (CI)** vetting incorporates **Polygraph** testing (**CI Poly**, **Lifestyle Poly**, **Full Scope Poly**) and **SEAD 3 Reporting** of **Foreign Contact** and **Foreign Travel**. Personnel are segregated into **Cleared** vs. **Uncleared** and **US Person (USP)** vs. **Foreign National (FN)** under **NISP**, **NISPOM**, **SAP NISPOM**, **ITAR**, **EAR**, **Export Control**, and **Supply Chain** (SCRM) statutes.

5.2 Physical Security (SCIF/SAPF), PACS/IDS, Continuity, and TEMPEST/EMSEC Matrix

Taxonomy Terms (Exact Match)	Primary Governing Authority	Engineering, Physical & Electromagnetic Specification
SCIF, SAPF, FACILITY, SECURE ROOM, VAULT,	ICD 705; IC Tech Spec v1.5.1; AA-F-358K; FF-L-	Every **SCIF**, **SAPF**, **Secure Room**, or reinforced concrete **Vault** inside a **Facility** requires formal **SCIF**

Taxonomy Terms (Exact Match)	Primary Governing Authority	Engineering, Physical & Electromagnetic Specification
SAFE, GSA APPROVED, CLASS 5, CLASS 6, LOCK, COMBINATION, KEY, CONTROLLED AREA, LIMITED AREA, EXCLUSION AREA, SECURE AREA, SCIF ACCREDITATION, SAPF ACCREDITATION, FACILITY ACCREDITATION, TEMPEST ACCREDITATION	2740B; DoDM 5200.08 Vol 3	**Accreditation**, **SAPF Accreditation**, **Facility Accreditation**, and **TEMPEST Accreditation** across concentric **Controlled Area**, **Limited Area**, and **Exclusion Area** / **Secure Area** perimeters. Closed storage uses a **GSA Approved Safe** (**Class 5** 10-min forced-entry or **Class 6** 20-hr surreptitious container) secured by an FF-L-2740B electromechanical **Lock** and **Combination** with strict administrative **Key** control.
PACS, PHYSICAL ACCESS CONTROL SYSTEM, ACS, ACCESS CONTROL, ACCESS CONTROL SYSTEM, BADGE, CARD READER, READER, BIOMETRIC, PIN, TURNSTILE, MANTRAP, AIRLOCK, ESCORT, VISITOR, ESCORT REQUIRED, UNESCORTED	FIPS 201-3; NIST SP 800-116 Rev. 1; IC Tech Spec v1.5.1 Ch. 8	The **Physical Access Control System (PACS / ACS / Access Control / Access Control System)** validates a cryptographic CAC/PIV **Badge** at an OSDP v2 **Card Reader** (**Reader**) paired with a **PIN** and/or **Biometric** scan across an optical **Turnstile** or interlocked **Mantrap** (**Airlock**). Every **Visitor** is verified for **Unescorted** access or assigned an indoctrinated **Escort** under **Escort Required** controls.
IDS, INTRUSION DETECTION, ALARM, CCTV, MOTION, DOOR CONTACT, GLASS BREAK, TAMPER, RESPONSE, GUARD, FORCE, ARMED, UNARMED, ROVING, FIXED, POST, ORDERS, SOP, EAP, EMERGENCY ACTION PLAN, CONTINUITY, COOP, DISASTER RECOVERY, BACKUP, OFFSITE, SECURE STORAGE, ARCHIVE, RETENTION	UL 2050; IC Tech Spec v1.5.1 Ch. 9; FCD 1/2; NIST SP 800-34; 36 CFR Part 1234	UL 2050 **Intrusion Detection System (IDS / Intrusion Detection / Alarm)** combines BMS **Door Contact**, dual-tech **Motion**, **Glass Break**, 24/7 **Tamper** switches, and perimeter **CCTV** to trigger a ≤ 5–15 min **Response** by an **Armed** or **Unarmed Guard Force** across **Fixed** and **Roving Post** assignments governed by Post **Orders**, **SOP**, and **Emergency Action Plan (EAP)**. **Continuity (COOP)** and **Disaster Recovery** replicate encrypted **Backup** datasets to **Offsite Secure Storage** and NARA **Archive** / **Retention** vaults.
TEMPEST, EMSEC, COMPROMISING EMANATIONS, COMPROMISING, EMANATIONS, RED/BLACK, RED/BLACK SEPARATION, RED, BLACK, RED NETWORK, BLACK NETWORK, PLAIN TEXT, PLAIN, CIPHER TEXT, CIPHER, SHIELDING, FILTERING, POWER FILTERING,	NSTISSAM TEMPEST/1-92; CNSSAM TEMPEST/01-13; MIL-STD-188-124B; MIL-PRF-15733J	**TEMPEST** / **EMSEC** suppresses **Compromising Emanations** (**Compromising** / **Emanations**) by enforcing **Red/Black Separation** (**Red/Black**) between **Red** / **Red Network Plain Text** (**Plain**) and **Black** / **Black Network Cipher Text** (**Cipher**) via ≥ 100 dB RF **Shielding**, MIL-PRF-15733 **Power Line Filter** (**Power Filtering** / **Filtering** / **Power Line** / **Power**) and **Data Line Filter** (**Data Line** / **Data**) units, ≤ 10 Ω equipotential **Grounding**, and all-dielectric **Fiber Optic** (**Fiber** / **Optical**) waveguided **Optical Isolation** (**Isolation**).

Taxonomy Terms (Exact Match)	Primary Governing Authority	Engineering, Physical & Electromagnetic Specification
POWER, POWER LINE, POWER LINE FILTER, DATA, DATA LINE, DATA LINE FILTER, GROUNDING, FIBER, FIBER OPTIC, OPTICAL, OPTICAL ISOLATION, ISOLATION		

5.3 Classified Networks, Cloud Impact Levels, PKI/HSM Identity, Cryptographic Systems, eBPF/XDP, and Sanitization Matrix

Taxonomy Terms (Exact Match)	Primary Governing Authority	Engineering, Cryptographic & Kernel/Sanitization Specification
CNSSI 1253, JWICS, NSANET, SIPRNET, SIPR, NIPRNET, NIPR, CLASSIFIED NETWORK, NETWORK, CLASSIFIED ENCLAVE, SECURE ENCLAVE, DoD CLOUD, SECURE CLOUD, COMMERCIAL CLOUD, IMPACT LEVEL, IL2, IL4, IL5, IL6, C2S, AWS SECRET, AWS TOP SECRET, AZURE GOVERNMENT, AZURE IL5, AZURE IL6, GOOGLE, ORACLE, BCAP, VDSS, VDMS, TCCM	CNSSI 1253; DISA CC SRG; DoD Cloud Playbook; AWS SCCA; Azure IL6; Google DISA; Oracle Defense	**CNSSI 1253** categorizes **JWICS** and **NSANet** (TS/SCI), **SIPRNet** (**SIPR**, Secret), and **NIPRNet** (**NIPR**, Unclass/CUI) across **Classified Network** (**Network**), **Classified Enclave**, and **Secure Enclave** boundaries. **DoD Cloud** / **Secure Cloud** / **Commercial Cloud** workloads scale across **Impact Level IL2**, **IL4**, **IL5**, **IL6**, and IC **C2S** across **AWS Secret**, **AWS Top Secret**, **Azure Government**, **Azure IL5**, **Azure IL6**, **Google**, and **Oracle**, protected by DISA's **BCAP**, **VDSS**, **VDMS**, and **TCCM** stack.
HSPD-12, FIPS 201, CAC, PIV, CAC/PKI, PKI, CERTIFICATE, PRIVATE KEY, TOKEN, HSM, COMSEC, TRANSEC, OPSEC, INFOSEC, COMSEC ACCOUNT, CRYPTO CUSTODIAN, KEY MANAGEMENT, KEYMAT, KEYING MATERIAL, CRYPTO, ENCRYPTION, CRYPTO MODULE, FIPS 140, FIPS 140-2, FIPS 140-3, COMMON CRITERIA, EAL, TYPE 1, TYPE 2, TYPE 3, TYPE 4, HIGH ASSURANCE, HAIPE, IPSEC, INLINE ENCRYPTOR, INLINE, NETWORK ENCRYPTOR, ENCRYPTOR, KG-175, TACLANE, FILL DEVICE,	HSPD-12; FIPS 201-3; FIPS 140-2/140-3; ISO 15408 CC; DoDI 8523.01; KG-175 TACLANE; TM 11-5810-292-13&P	**HSPD-12** and **FIPS 201** bind **CAC** and **PIV** smartcard **Token** credentials into **CAC/PKI** and DoD **PKI** using on-card **Private Key** slots, X.509 **Certificate** chains, and **HSM** roots. Under **INFOSEC**, **OPSEC**, **COMSEC**, and **TRANSEC**, each **COMSEC Account** and **Crypto Custodian** governs **Key Management** (**Management**) and **KEYMAT** (**Keying Material**) for **Crypto** / **Encryption** across **FIPS 140** (**FIPS 140-2** / **FIPS 140-3**) **Crypto Module** units, **Common Criteria EAL**, and NSA **Type 1**, **Type 2**, **Type 3**, and **Type 4** products. **High Assurance HAIPE Inline Encryptor** (**Inline** / **Network Encryptor** / **Encryptor**) units—specifically the **KG-175 TACLANE**—encapsulate Red traffic in **IPsec** ESP tunnels keyed by a **Fill Device** (**SKL**, **AN/CYZ-10**, **KOI-18**).

Taxonomy Terms (Exact Match)	Primary Governing Authority	Engineering, Cryptographic & Kernel/Sanitization Specification
SKL, AN/CYZ-10, KOI-18, MANAGEMENT		
eBPF, BPF, XDP, SECURE TELEMETRY, MALWARE DETECTION, ANTI-FORENSIC, DIGITAL FORENSICS, FORENSICS, EVIDENCE, CHAIN OF CUSTODY, CoC, MEDIA SANITIZATION, SANITIZATION, DESTRUCTION, DISPOSAL, SECURE DISPOSAL, NIST 800-88, CLEAR, PURGE, DESTROY, CRYPTOGRAPHIC ERASE, KEY DESTRUCTION, ZEROIZATION, SECURE ERASE, ATA SECURE ERASE, NVME FORMAT, OVERWRITE, DOD 5220.22-M, GUTMANN, SCHNEIER, RANDOM, VERIFY, SHREDDING, SHREDDER, CROSS-CUT, BURNING, INCINERATOR, PULPING, DEGAUSSER, NSA APPROVED, CERTIFICATE OF DESTRUCTION, COD, MEDIA, HARD DRIVE, SSD, TAPE, OPTICAL, PAPER, DOCUMENT	RFC 9669; Linux eBPF/XDP; NIST SP 800-86; NIST SP 800-88 Rev. 1/2; NSA/CSS Manual 9-12	In-kernel **eBPF** / **BPF** and driver-level **XDP** provide verifiable **Secure Telemetry**, **Malware Detection**, and **Anti-Forensic** blocking, feeding **Digital Forensics** (**Forensics**) **Evidence** preserved under an immutable **Chain of Custody (CoC)**. Terminal **Media Sanitization** (**Sanitization**), **Destruction**, and **Secure Disposal** (replacing raw **Disposal**) under **NIST 800-88** apply **Clear** (**Overwrite** via **Random**, **DoD 5220.22-M**, **Schneier**, or **Gutmann** + **Verify**), **Purge** (**ATA Secure Erase** / **Secure Erase**, **NVMe Format**, **Cryptographic Erase** / **Key Destruction** / **Zeroization**, or **NSA Approved Degausser**), and **Destroy** (**Cross-Cut Shredding** in a ≤ 1 × 5 mm **Shredder**, ≤ 2 mm disintegration, **Burning** in an **Incinerator**, or **Pulping**) across **Media** (**Hard Drive**, **SSD**, **Tape**, **Optical**, **Paper**, **Document**), documented via a **Certificate of Destruction (COD)**.

6. Categorized Primary-Source Archival Bibliography

- **Statutory & Executive Orders (U.S. Code, National Archives ISOO & eCFR):** Executive Order 13526 (Classified National Security Information, 75 FR 707); 32 CFR Part 2001 (ISOO Implementing Directive); 10 U.S.C. § 119 (Special Access Programs: Congressional Oversight); 50 U.S.C. § 3341 (Security Clearances & Continuous Vetting); 32 CFR Part 117 (National Industrial Security Program Operating Manual — NISPOM Rule); 22 CFR Parts 120–130 (International Traffic in Arms Regulations — ITAR); 15 CFR Parts 730–774 (Export Administration Regulations — EAR); 36 CFR Part 1234 (NARA Electronic & Archival Records Management); HSPD-12 (Policy for a Common Identification Standard for Federal Employees and Contractors).
- **Department of Defense Directives, Instructions & Manuals (DoD Issuances / WHS):** DoDD 5205.07 (Special Access Program Policy); DoDM 5205.07 (SAP Security Manual) & DoDM 5205.07 Vol 4 (SAP Marking); DoDI 5205.11 (DoD SAP Administration); DoDM 5200.01 Vols 1–3 (DoD Information Security Program); DoDM 5200.02 (DoD Personnel Security Program); DoDM 5200.45 (Instructions for Developing Security Classification Guides); DoDI 5200.39 (Critical Program Information Identification and Protection Within RDT&E); DoDM 5105.21 Vols 1–3 (Sensitive Compartmented Information Administrative Security

Manual); DoDI 8520.02 (Public Key Infrastructure and Public Key Enabling); DoDI 8523.01 (Communications Security — COMSEC).

- **Intelligence Community Directives, SEADs & Physical/TEMPEST Engineering Standards:** ICD 704 (Personnel Security Standards for Access to SCI); ICD 705 (Sensitive Compartmented Information Facilities); IC Technical Specifications for Construction and Management of SCIFs v1.5.1; SEAD 3 (Reporting Requirements); SEAD 4 (National Security Adjudicative Guidelines); SEAD 6 (Continuous Evaluation); SEAD 7 (Reciprocity); AA-F-358K / FED-STD-809D (GSA Class 5 & 6 Security Containers); FF-L-2740B (Electromechanical Combination Locks); UL 2050 (National Industrial Security Systems); NSTISSAM TEMPEST/1-92 & CNSSAM TEMPEST/01-13 (Red/Black Installation Guidance); MIL-STD-188-124B (Grounding, Bonding and Shielding); MIL-PRF-15733J (RF Interference Filters).
- **CNSS, DISA, NIST, NSA & Kernel/Sanitization Specifications:** CNSSI 1253 (Security Categorization and Control Selection for NSS); CNSSI 4009 (Committee on National Security Systems Glossary); DISA Cloud Computing Security Requirements Guide (CC SRG); DoD CIO Cloud Security Playbook Vol 1; NIST FIPS 201-3; NIST FIPS 140-3; NSA CNSA 2.0 Algorithm Suite; General Dynamics KG-175 TACLANE HAIPE Encryptors; Army TM 11-5810-292-13&P (KOI-18, AN/CYZ-10, SKL Fill Devices); IETF RFC 9669 (BPF Instruction Set Architecture); Linux Kernel eBPF Verifier, LSM BPF & AF_XDP; NIST SP 800-86 (Forensic Techniques); NIST SP 800-88 Rev. 1 & Rev. 2 (Guidelines for Media Sanitization); NSA/CSS Policy Manual 9-12 & Evaluated Products Lists (EPL).